

Algorithms 2

Lovasz Local Lemma

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1. Lovasz Local Lemma

Definition 1 The scenario:

Let $E_1, \dots, E_n$ be bad events (over the same sample space).

We wish to prove: $P\left[\bigcap_{i=1}^n \overline{E_i}\right] > 0$ – no bad event occurs.

- If $E_1, \dots, E_n$ are pairwise independent, then we can argue as follows:
 - Recall: events A and B are independent if $P[A \cap B] = P[A]P[B]$.
 - If A, B are independent, then so are $\overline{A}, \overline{B}$:

$$\begin{aligned} P[\overline{A} \cap \overline{B}] &= 1 - P[A \cup B] \\ &= 1 - P[A] - P[B] + P[A \cap B] \\ &= (1 - P[A])(1 - P[B]) \\ &= P[\overline{A}]P[\overline{B}] \end{aligned}$$

- Back to our problem:
- $E_1, \dots, E_n$ pairwise independent $\Rightarrow \overline{E_1}, \dots, \overline{E_n}$ pairwise independent.
- If $P[E_i] < 1$ for all $i \in [n]$, then:

$$P\left[\bigcap_{i=1}^n \overline{E_i}\right] = \prod_{i=1}^n P[\overline{E_i}] = \prod_{i=1}^n (1 - P[E_i]) > 0$$

(assuming this equality holds).

- But pairwise independence does **not** imply

$$P\left[\bigcap_{i=1}^n \overline{E_i}\right] = \prod_{i=1}^n P[\overline{E_i}].$$

- Mutual independence does.
- What if $E_1, \dots, E_n$ do depend on one another in some way?
- The Lovasz local lemma provides a partial solution.

Definition 2 Events $E_1, \dots, E_n$ are said to be **mutually independent** if

$$P\left[\bigcap_{i \in I} E_i\right] = \prod_{i \in I} P[E_i]$$

whenever $\emptyset \neq I \subseteq [n]$.

Definition 3 (Alternative definition) Event A is mutually independent of events $E_1, \dots, E_n$ if

$$P\left[A \mid \bigcap_{i=1}^n B_i\right] = P[A]$$

where $B_i \in \{E_i, \overline{E_i}\}$ for all $i \in [n]$.

- No matter whether each E_i occurs or not, this does not affect $P[A]$.
- For a proof of equivalence, see the booklet.

- Mutual independence $\Rightarrow$ pairwise independence.
- The converse implication fails.

Example. Let $\Omega := \{1, 2, 3, 4\}$ with the discrete uniform measure. $A := \{1, 2\}$, $B := \{1, 3\}$, $C := \{2, 3\}$.

- $P[A] = P[B] = P[C] = \frac{1}{2}$.
- A, B are independent: $P[A \cap B] = P[\{1\}] = \frac{1}{4} = P[A]P[B]$.
- The same applies to B, C and A, C .
- Yet $P[A \cap B \cap C] = P[\emptyset] = 0 \neq P[A]P[B]P[C]$.
- So A, B, C are pairwise independent but **not** mutually independent.

- Mutual independence is a **stronger** form of independence than pairwise independence.
- We have seen that $E_1, \dots, E_n$ pairwise independent $\Rightarrow \overline{E_1}, \dots, \overline{E_n}$ pairwise independent.
- This property is also supported in mutual independence:

Theorem 1 If $E_1, \dots, E_n$ are mutually independent, then so are $\overline{E_1}, \dots, \overline{E_n}$.

For a proof, see the booklet.

- The following is a handy tool for determining mutual independence.

Theorem 2 (Mutual independence principal)

Let $X := (X_1, \dots, X_m)$ be a sequence of pairwise independent trials. Let $A_1, \dots, A_n$ be events, and for each $i \in [n]$ let A_i be determined by trials $F_i \subseteq X$. Given $I \subseteq [n]$ and $j \in [n] \setminus I$, if

$$F_j \cap \left(\bigcup_{i \in I} F_i \right) = \emptyset,$$

then A_j is mutually independent of $(A_i)_{i \in I}$.

- There are several versions. We state one of those.
- The version stated here is called the **symmetric version**.
- We focus on applications of the lemma and not its proof(s).

The dependency graph

Definition 4 Events $E_1, \dots, E_n$. A dependency graph $D := D(E_1, \dots, E_n)$ is any graph whose vertex set is $E_1, \dots, E_n$ and supporting the property: for every $i \in [n]$, event E_i is mutually independent of the events $\{E_j : j \neq i \wedge \{i, j\} \notin E(D)\}$.

Remark

Sometimes a directed version of such graphs is more useful (i.e., in **asymmetric** applications).

Remark

Danger: We did **not** just say that we build a graph on $E_1, \dots, E_n$ by placing an edge (E_i, E_j) if E_i, E_j are dependent.

Theorem 3 (The symmetric local lemma)

Let $E_1, \dots, E_n$ be events such that the following holds:

1. **(Symmetry)** $P[E_i] \leq p$ for all $i \in [n]$.
2. **(Limited dependency)** $\Delta(D(E_1, \dots, E_n)) \leq d$.
3. **(Bound p and d)** $e \cdot p \cdot (d + 1) \leq 1$ (up to a constant).

Then $P\left[\bigcap_{i=1}^n \overline{E_i}\right] > 0$.